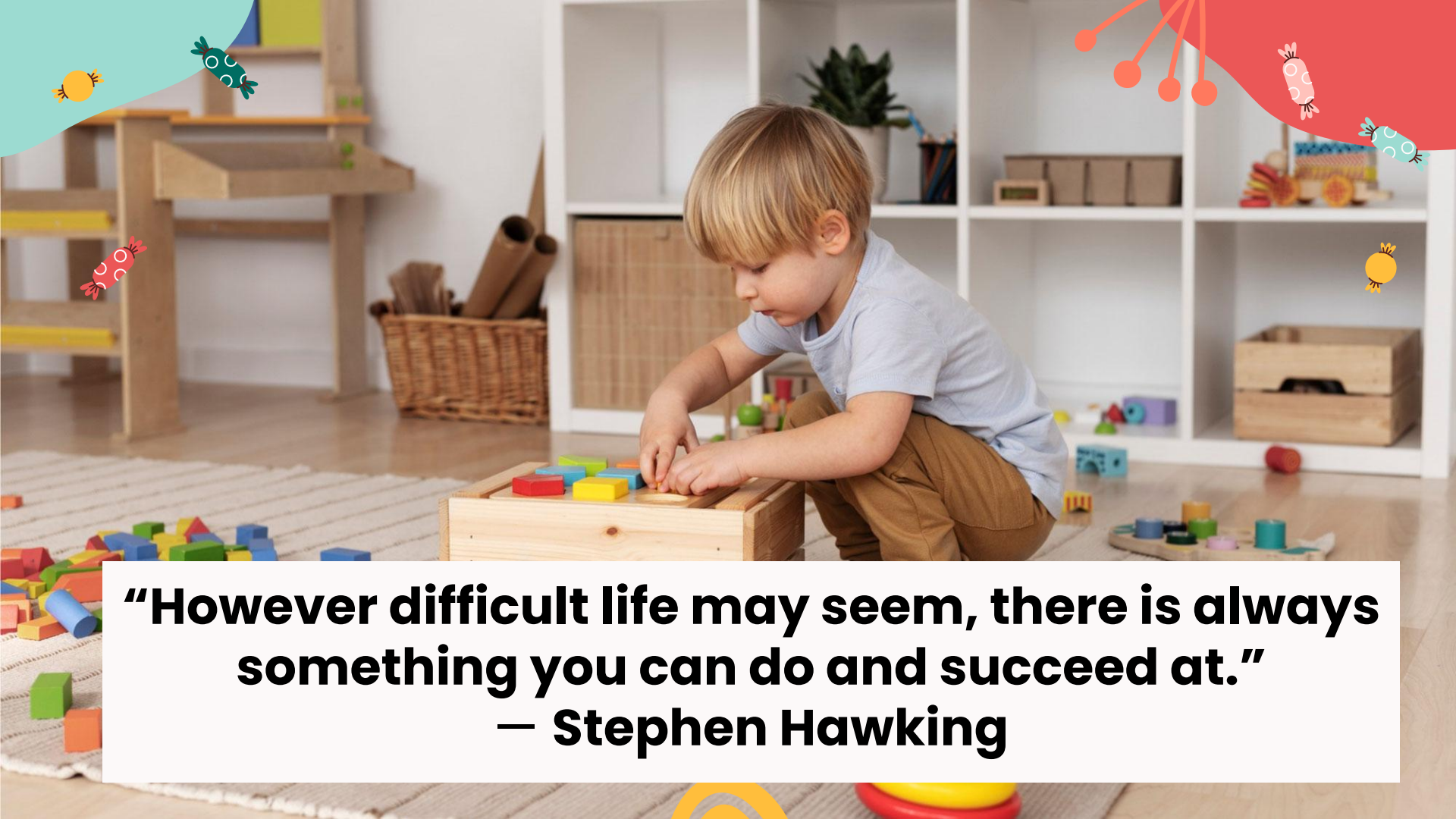


ABILITY PLAY : EVERY MOMENT MATTERS

Democratizing Pediatric Rehab Through Adaptive Sport for KIDS

THERAPY TODAY STRONGER TOMORROW





**“However difficult life may seem, there is always something you can do and succeed at.”
— Stephen Hawking**

THE PROBLEM

Their bodies move differently. Their childhood shouldn't feel different

HOW BIG IS THE NEED?

~1 in 250–1,000 children are affected by cerebral palsy globally, depending on the population studied.

~3 per 1,000 live births is a commonly cited estimate for CP prevalence in India.



KIDS WITH CP CAN'T PLAY INDEPENDENTLY

Children with cerebral palsy often have difficulties with **balance, muscle control, coordination, and posture**, making many everyday play activities difficult to access without assistance.

MISS OUT ON PLAYING WITH PEERS

When activities require movements they cannot easily perform, CP children are often **left out of spontaneous play** with other children — turning play into an experience of watching rather than participating.

THEIR ABILITIES AREN'T THE LIMIT — THE PLAY EXPERIENCE IS

Most play activities are designed around a **single way of moving**. They don't adapt to different levels of mobility and motor control, preventing children with CP from **actively participating, exploring, and playing alongside their peers**.

OUR SOLUTION

Exercise-to-Sport Conversion:

Given Maps prescribed therapy movements to meaningful in-game sports actions. Turns repetitive exercises into activities such as basketball, football, racing, and other interactive sports.



Personalized Adaptation: Adjusts difficulty based on each child's individual mobility baseline rather than using a one-size-fits-all approach.

Performance Tracking: Records movement performance, repetitions, accuracy, range, timing, and progression to help track rehabilitation over time.



Progressive Rehabilitation:

Automatically increases or decreases challenge as the child's movement ability changes.

Device-Agnostic: Designed to work with commonly available hardware such as a **phone/tablet camera, webcam, motion sensors, or other compatible input devices**, reducing dependence on specialized rehabilitation equipment.

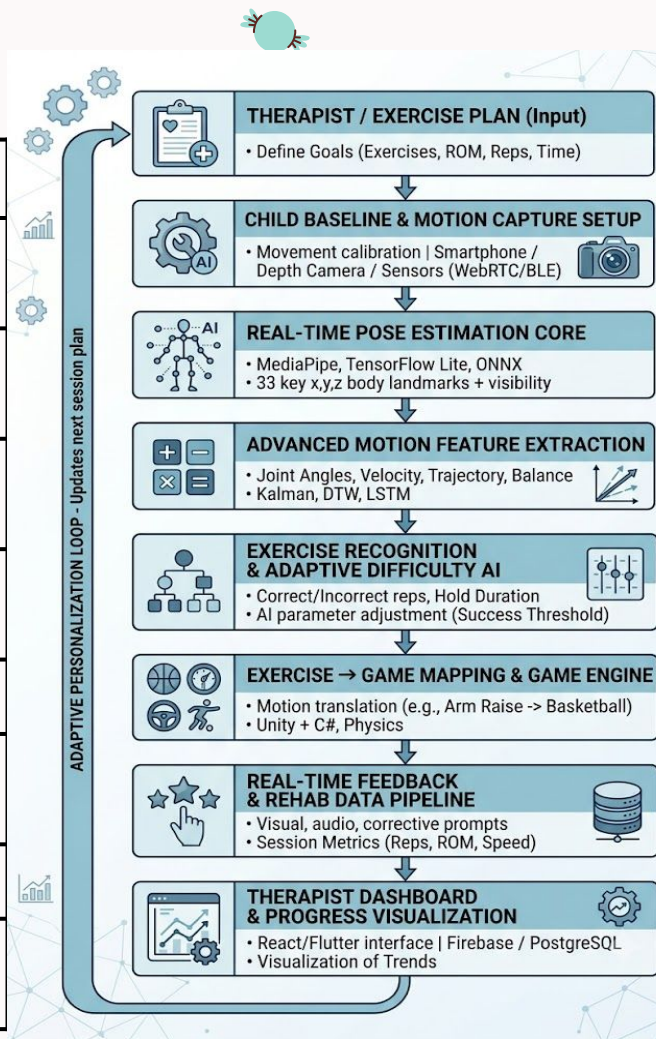
Tracks the child's physical movements using everyday devices.

Converts therapy exercises into fun virtual sports like basketball.

Adapts game difficulty and targets based on the child's ability and progress.

TECHNICAL APPROACH

What We Need	Technology	Purpose
Movement Detection	MediaPipe	Detect body joints
Movement Analysis	OpenCV + Joint Mathematics	Calculate ROM, angles & speed
Exercise Recognition	LSTM / TCN + DTW	Identify and evaluate exercises
Game Conversion	Unity + C#	Convert exercises into sports actions
Personalization	Adaptive AI	Adjust difficulty to the child
Real-Time Interaction	WebRTC	Connect movement with gameplay
Data Storage	Firebase	Store rehabilitation data
Therapist Monitoring	React / Flutter	Track progress and performance



CASE STUDY: ADAPTIVE BASKETBALL



Child's arm movement
controls the game



Target height & distance
adapt to the child's reach



Successful movement
immediately triggers a virtual
shot

AbilityPlay - Adaptive Basketball Movement framework for Rehabilitation



DASHBOARD PREVIEW

ADAPTIVE REHABILITATION / CASE STUDY 001

Play is a measurable motion.

AbilityPlay – Gamified Pediatric Rehabilitation. Adaptive Basketball turns body movement into game control, so a therapist can see the signal and a child can feel the win.

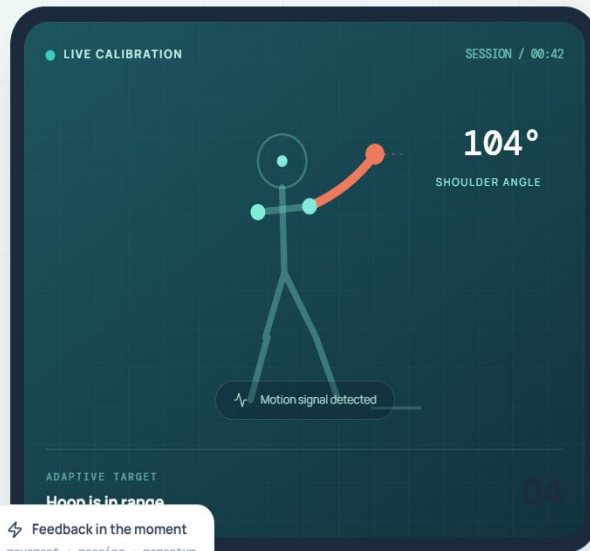
Explore the simulator →

See how it works ↘

33 KEYPOINTS
TRACKED

<50ms REAL-TIME
ADAPTABILITY

100% DYNAMIC
SCALING



DEMO :

https://youtu.be/HwrwKg5DfGg?si=WDAKc_gs

DRIVE :

<https://drive.google.com/file/d/1TsAHvaRc47qjPvXQi4qNEw/view?usp=sharing>

GITHUB :

<https://github.com/divyadhasini/AbilityPlay>

INFERENCES



Child Profile	Current Physical Movement	Real-Time Game Adaptation	Therapeutic Objective 
Child A – Beginner 	Limited arm reach / severe restriction	Larger, lower basketball target	Encourage motor activation & movement confidence
Child B – Progressing	Moderate arm reach	Medium target + moderate game speed	Build endurance & expand active ROM
Child C – Advanced	Higher arm reach	Smaller, farther target + higher speed	Improve accuracy, speed & maximum reach

- ❖ The system analyzes each child's current movement ability. Target height, size, and distance are adjusted accordingly.
- ❖ This ensures the child can participate without unnecessary difficulty.
- ❖ Targets remain challenging but achievable for each child.

- ❖ Game difficulty changes based on the child's performance and progress.
- ❖ Speed, movement range, repetitions, and success thresholds are adjusted.
- ❖ As the child improves, the challenge gradually increases.



IMPACTS AND BENEFITS



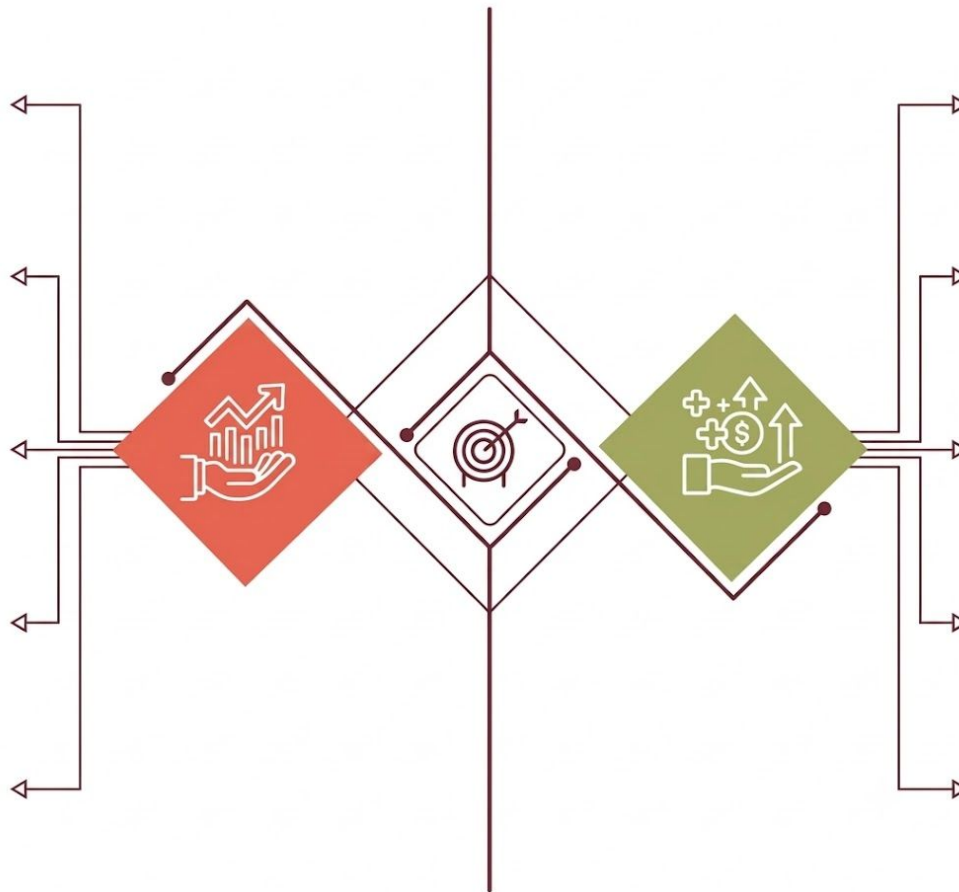
Participation :Increases children's motivation and participation in regular rehabilitation.

Parameters : Improves mobility, coordination, balance, and range of motion through active gameplay.

Customised : Provides personalized therapy that adapts to each child's abilities and progress.

Engaging : Extends rehabilitation beyond clinics by enabling engaging home-based therapy.

Progress : Creates measurable progress by continuously tracking movement and rehabilitation performance.



Fun & Motivating – Makes rehabilitation feel like play rather than exercise.

Accessible – Works with everyday hardware, making home therapy more practical.

Adaptive – Automatically adjusts targets, speed, and difficulty to the child.

Measurable – Provides objective performance data for monitoring improvement.

Inclusive & Confidence-Building – Creates achievable challenges that encourage children to move and participate.



ROADMAP

Prototype Development –

Build core motion tracking and convert basic therapy exercises into interactive sports games.



Personalization Engine –

Develop adaptive algorithms to customize targets, difficulty, and therapy goals for each child.



Pilot Testing – Test the platform with children and therapists to validate usability, engagement, and rehabilitation outcomes.

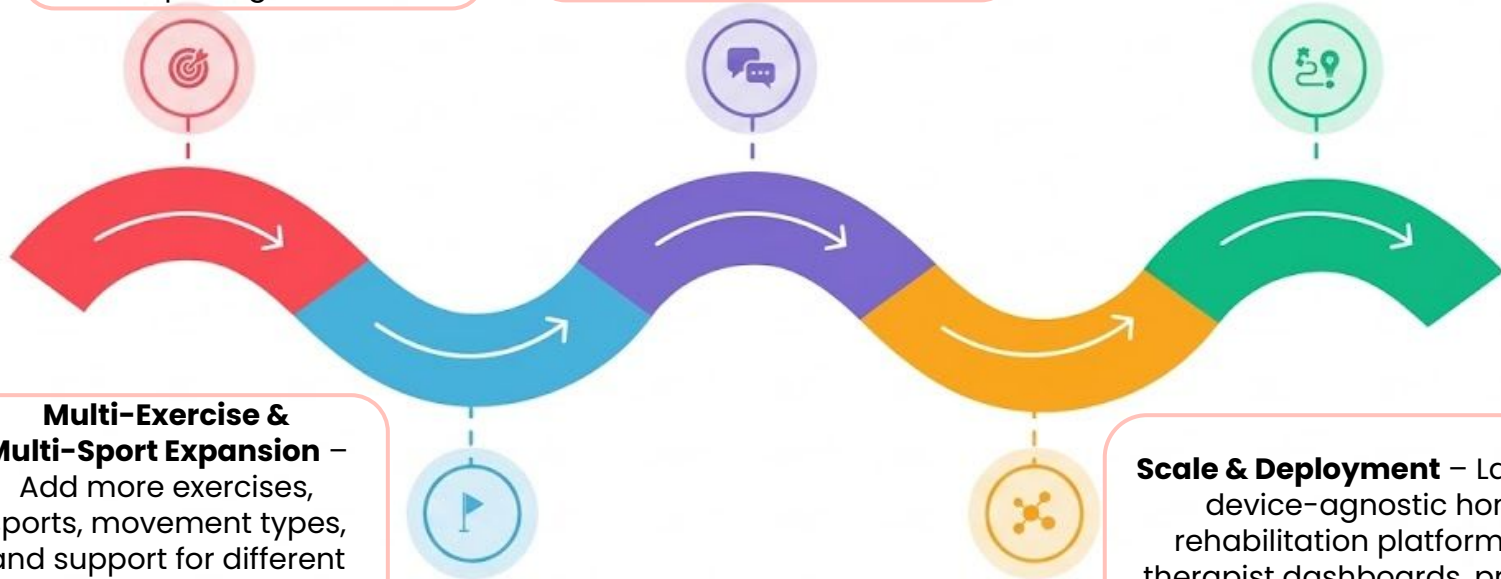


Multi-Exercise & Multi-Sport Expansion –

Add more exercises, sports, movement types, and support for different mobility levels.



Scale & Deployment – Launch a device-agnostic home rehabilitation platform with therapist dashboards, progress tracking, and wider accessibility.



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Thanks!



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